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# **A PROJECT REPORT ON**

**HR Analytics: Employee Attrition Analysis**

**DATA ANALYST INTERNSHIP PROGRAM  
UNIFIED MENTOR**

**Submitted By**

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# ***DECLARATION***

I, Prasannajit Biswal, hereby declare that  
the project titled  
"HR Analytics Dashboard Using MySQL &  
Power BI"  
submitted as part of the Data Analyst  
Internship Program  
at Unified Mentor is my original work.

***Prasannajit Biswal***

# ***ACKNOWLEDGEMENT***

I would like to express my sincere gratitude to Unified Mentor for providing me with the opportunity to participate in the Data Analyst Internship Program

This project, "HR Analytics Dashboard Using MySQL & Power BI", has helped me gain practical knowledge and hands-on experience in data analysis, database management, data visualization, and business intelligence tools.

I am thankful to the mentors and support team at Unified Mentor for their guidance, encouragement, and valuable suggestions throughout the internship. Their support played an important role in the successful completion of this project.

I also extend my appreciation to everyone who directly or indirectly contributed to the completion of this project.

# ***ABSTRACT***

The HR Analytics Dashboard Using MySQL & Power BI project was developed to analyse employee data and gain insights into workforce trends and employee attrition. The project uses employee information such as age, gender, department, salary, overtime status, and attrition records to identify patterns that can support human resource decision-making.

The dataset was managed using MySQL (XAMPP) and analysed in Power BI. Various DAX measures and interactive visualizations were created to monitor key performance indicators, including total employees, attrition count, and attrition rate. The dashboard also provides analysis based on department, gender, overtime, salary, and age distribution.

The project helps organizations understand factors affecting employee retention and supports data-driven workforce planning. The dashboard presents information in a clear and interactive format, making it easier for decision-makers to identify trends and take appropriate actions.

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# ***INTRODUCTION***

The HR Analytics Dashboard project was developed to analyze employee-related data and generate meaningful insights for workforce management. Human Resource (HR) analytics plays an important role in understanding employee behavior, identifying attrition patterns, and supporting organizational decision-making.

In this project, employee data from Green Destinations was analyzed using MySQL and Power BI. The dashboard was designed to provide a clear view of employee demographics, attrition trends, salary distribution, overtime impact, and departmental performance.

The project aims to transform raw employee data into useful business insights through interactive visualizations and key performance indicators (KPIs). These insights can help organizations improve employee retention, workforce planning, and overall HR effectiveness.

# ***OBJECTIVES***

The main objectives of this project are:

- To analyze employee data using MySQL and Power BI.
- To identify employee attrition trends across different departments.
- To calculate key HR metrics such as Total Employees, Attrition Count, and Attrition Rate.
- To study the impact of factors such as gender, overtime, age, and job role on employee attrition.
- To create an interactive dashboard for HR analysis and decision-making.
- To present workforce insights through effective visualizations and KPI indicators.

# ***TOOLS USED***

## **1. MySQL (XAMPP)**

MySQL was used for storing and managing the employee dataset. XAMPP provided a local environment to create, import, and manage the HR database using phpMyAdmin.

## **2. Power BI Desktop**

Power BI Desktop was used to create interactive dashboards, charts, KPI cards, and visual reports for analyzing employee attrition and workforce trends.

## **3. DAX (Data Analysis Expressions)**

DAX was used to create measures such as Total Employees, Attrition Count, and Attrition Rate for dynamic calculations and dashboard analysis.

# ***DATASET DESCRIPTION***

The dataset used in this project is the Green Destinations HR Employee Dataset. It contains information about employees, including age, gender, department, job role, salary, overtime status, education level, and attrition details.

## Dataset Information

- Total Employees : 1470
- Company : Green Destinations
- Data Type : HR Employee Data

## Key Fields

- Employee ID
- Age
- Gender
- Department
- Job Role

- Monthly Income
- Overtime
- Attrition
- Education
- Marital Status

The dataset was used to identify employee attrition trends, workforce demographics, and factors influencing employee retention.

# PROJECT WORKFLOW

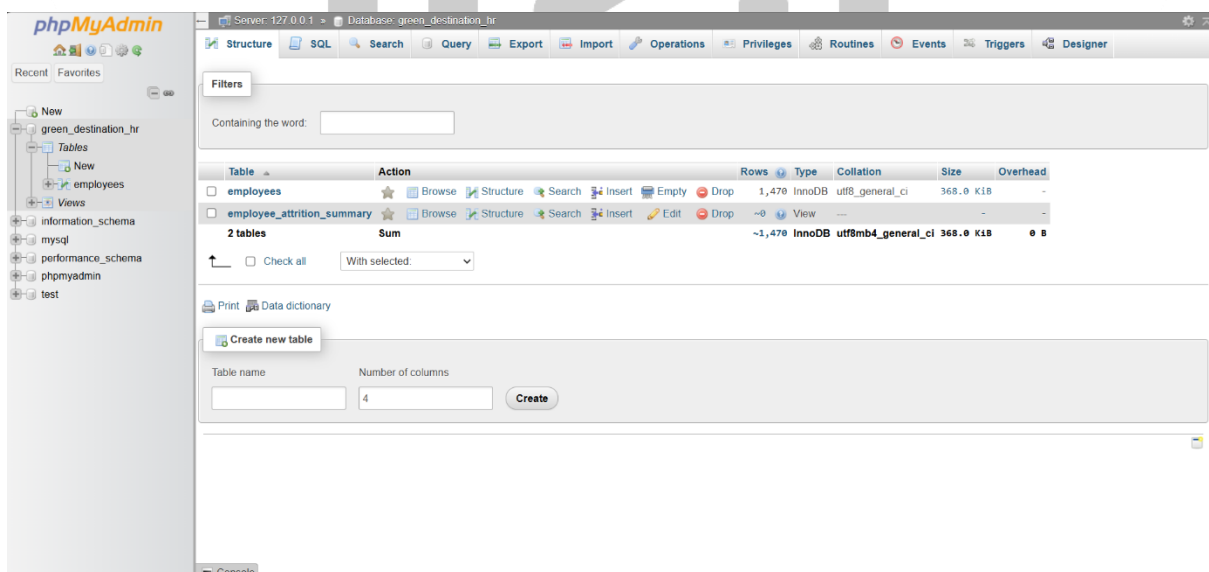
The HR Analytics Dashboard project was completed through the following steps:

## Step 1: Data Collection

The HR employee dataset of Green Destinations was collected for analysis. The dataset contained employee information such as age, gender, department, salary, overtime status, and attrition.

## Step 2: Database Creation Using MySQL

The dataset was imported into MySQL using XAMPP and managed through phpMyAdmin. The employee data was stored in database tables for analysis.



### Step 3: Data Cleaning & Validation

After importing the dataset into MySQL, data quality checks were performed to ensure the accuracy and reliability of the employee data. Missing values, duplicate employee records, and invalid entries were checked using SQL queries. Key fields such as Employee Number, Attrition, Age, Department, and Monthly Income were validated. The dataset was confirmed to be suitable for further analysis and reporting.

### Step 4: Data Import into Power BI

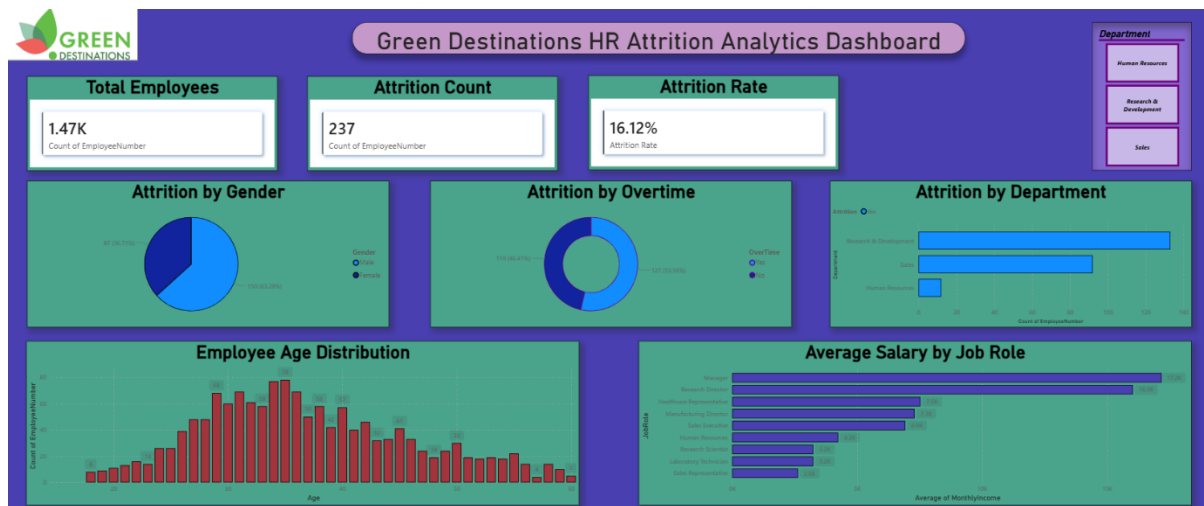
The employee data was imported into Power BI for visualization and reporting purposes.

### Step 5: Data Analysis and DAX Measures

Several DAX measures were created, including Total Employees, Attrition Count, and Attrition Rate to support dashboard analysis.

## Step 6: Dashboard Development

Interactive charts, KPI cards, and slicers were designed in Power BI to analyze workforce trends and employee attrition.



## Step 7: Insight Generation

The dashboard was used to identify attrition patterns, departmental trends, overtime impact, and workforce demographics.

# ***DASHBOARD FEATURES***

The HR Analytics Dashboard was developed using Power BI to analyze employee data and provide meaningful insights through interactive visualizations and KPI indicators.

## Features Included in the Dashboard

### 1. KPI Cards

- Total Employees
- Attrition Count
- Attrition Rate

### 2. Attrition Analysis

- Attrition by Department
- Attrition by Gender
- Attrition by Overtime

### 3. Employee Demographics

- Age Distribution

### 4. Salary Analysis

- Average Salary by Job Role

### 5. Interactive Filtering

- Department Slicer

The dashboard enables users to explore employee trends, compare workforce metrics, and identify factors influencing employee attrition.

# ***KEY INSIGHTS***

The analysis of the HR employee dataset revealed several important insights:

- Employee attrition was observed across multiple departments, indicating the need for retention strategies.
- Employees working overtime showed higher attrition compared to employees who did not work overtime.
- Attrition patterns differed between male and female employees.
- Salary levels varied among different job roles, highlighting differences in compensation.
- The workforce was concentrated within specific age groups, providing useful demographic information for HR planning.

These insights can help organizations improve employee retention and make better workforce management decisions.



# ***CHALLENGES FACED***

During the development of this project, several challenges were encountered:

- Difficulty connecting MySQL directly with Power BI due to the absence of the MySQL connector.
- Understanding and creating DAX measures for dashboard calculations.
- Designing an effective dashboard layout while maintaining clarity and readability.
- Ensuring accurate data representation through charts, KPI cards, and visualizations.

These challenges were addressed through research, practice, and the effective use of Power BI and MySQL tools.

# **CONCLUSION**

The HR Analytics Dashboard Using MySQL and Power BI successfully transformed employee data into meaningful business insights through interactive visualizations and KPI indicators.

The project enabled the analysis of employee attrition, workforce demographics, salary distribution, and overtime impact. By utilizing MySQL for data management and Power BI for data visualization, valuable insights were generated to support HR decision-making.

This project enhanced practical skills in database management, data analysis, dashboard development, and business intelligence. It provided hands-on experience in using MySQL, DAX, and Power BI to solve real-world business problems and present data effectively.